Problem Set 13
Due: May. 31, 2021

1. Section 5.4 Problem 9.
   During each day, the probability that your computer's operating system crashes at least once is 5%, independent of every other day. You are interested in the probability of at least 45 crash-free days out of the next 50 days.

   (a) Find the probability of interest by using the normal approximation to the binomial.

   (b) Repeat part (a), this time using the Poisson approximation to the binomial.

   Solution:

   (a) Let S be the number of crash-free days, which is a binomial random variable with parameters $n = 50$ and $p = 0.95$, so that $\mathbb{E}[X] = 50 \cdot 0.95 = 47.5$ and $\sigma_S = \sqrt{50 \cdot 0.95 \cdot 0.05} = 1.54$. Using the normal approximation to the binomial, we find

   $$P(S \geq 45) = P\left(\frac{S - 47.5}{1.54} \geq \frac{45 - 47.5}{1.54}\right) \approx 1 - \Phi(-1.62) = \Phi(1.62) = 0.9474.$$

   A better approximation can be obtained by using the de Moivre-Laplace approximation, which yields

   $$P(S \geq 45) = P(S > 44.5) = P\left(\frac{S - 47.5}{1.54} \geq \frac{44.5 - 47.5}{1.54}\right) \approx 1 - \Phi(-1.95) = \Phi(1.95) = 0.9744.$$

   (b) The random variable S is binomial with parameter $p = 0.95$. However, the random variable $50 - S$ (the number of crashes) is also binomial with parameter $p = 0.05$. Since the Poisson approximation is exact in the limit of small p and large n, it will give more accurate results if applied to $50 - S$. We will therefore approximate $50 - S$ by a Poisson random variable with parameter $\lambda = 50 \cdot 0.05 = 2.5$. Thus,

   $$\begin{aligned} P(S \geq 45) &= P(50 - S \leq 5) \\ &= \sum_{k=0}^{5} P(n - S = k) \\ &= \sum_{k=0}^{5} e^{-\lambda} \frac{\lambda^k}{k!} \\ &= 0.958. \end{aligned}$$

   It is instructive to compare with the exact probability which is

   $$\sum_{k=0}^{5} \binom{50}{k} 0.05^k \cdot 0.95^{50-k} = 0.962.$$

Thus, the Poisson approximation is closer. This is consistent with the intuition that the normal approximation to the binomial works well when p is close to 0.5 or n is very large, which is not the case here. On the other hand, the calculations based on the normal approximation are generally less tedious.

2. Section 5.4 Problem 10.

   A factory produces X_n gadgets on day n, where the X_n are independent and identically distributed random variables, with mean 5 and variance 9.

   (a) Find an approximation to the probability that the total number of gadgets produced in 100 days is less than 440.

   (b) Find (approximately) the largest value of n such that

   $$\mathbf{P}(X_1 + \cdots + X_n \geq 200 + 5n) \leq 0.05$$

   (c) Let N be the first day on which the total number of gadgets produced exceeds 1000. Calculate an approximation to the probability that $N \geq 220$.

   Solution:

   (a) Let $S_n = X_1 + \cdots + X_n$ be the total number of gadgets produced in n days. Note that the mean, variance, and standard deviation of S_n is $5n$, $9n$, and $3\sqrt{n}$, respectively. Thus,

   $$\begin{aligned} P(S_{100} < 440) &= P(S_{100} \leq 439.5) \\ &= P\left(\frac{S_{100} - 500}{30} \leq \frac{439.5 - 500}{30}\right) \\ &\approx \Phi\left(\frac{439.5 - 500}{30}\right) \\ &= \Phi(-2.02) \\ &= 0.0217. \end{aligned}$$

   (b) The requirement $P(S_n \geq 200 + 5n) \leq 0.05$ translates to

   $$P\left(\frac{S_n - 5n}{3\sqrt{n}} \geq \frac{200}{3\sqrt{n}}\right) \leq 0.05.$$

   or, using a normal approximation,

   $$1 - \Phi\left(\frac{200}{3\sqrt{n}}\right) \leq 0.05,$$

   and,

   $$\Phi\left(\frac{200}{3\sqrt{n}}\right) \geq 0.95.$$

From the normal tables, we obtain $\Phi(1.65) \approx 0.95$, and therefore,

$$\frac{200}{3\sqrt{n}} \geq 1.65,$$

which finally yields $n \leq 1632$.

(c) The event $N \geq 220$ (it takes at least 220 days to exceed 1000 gadgets) is the same as the event $S_{219} \leq 1000$ (no more than 1000 gadgets produced in the first 219 days). Thus,

$$\begin{aligned}
P(N \geq 220) &= P(S_{219} \leq 1000) \\
&= P\left(\frac{S_{219} - 5 \cdot 219}{3\sqrt{219}} \leq \frac{1000 - 5 \cdot 219}{3\sqrt{219}}\right) \\
&= 1 - \Phi(2.14) \\
&= 0.0162.
\end{aligned}$$

3. Section 5.4 Problem 11.

Let $X_1, Y_1, X_2, Y_2, \ldots$ be independent random variables, uniformly distributed in the unit interval $[0, 1]$, and let

$$W = \frac{(X_1 + \cdots + X_{16}) - (Y_1 + \cdots + Y_{16})}{16}$$

Find a numerical approximation to the quantity

$$\mathbf{P}(|W - \mathbf{E}[W]| < 0.001)$$

Solution: Note that W is the sample mean of 16 independent identically distributed random variables of the form $X_i - Y_i$, and a normal approximation is appropriate. The random variables $X_i - Y_i$ have zero mean, and variance equal to $2/12$. Therefore, the mean of W is zero, and its variance is $(2/12)/16 = 1/96$. Thus,

$$\begin{aligned}
P(|W| < 0.001) &= P\left(\frac{|W|}{\sqrt{1/96}} < \frac{0.001}{\sqrt{1/96}}\right) \approx \Phi(0.001\sqrt{96}) - \Phi(-0.001\sqrt{96}) \\
&= 2\Phi(0.001\sqrt{96}) - 1 \approx 0.008.
\end{aligned}$$

Let us also point out a somewhat different approach that bypasses the need for the normal table. Let Z be a normal random variable with zero mean and standard deviation equal to $1/\sqrt{96}$. The standard deviation of Z, which is about 0.1, is much larger than 0.001. Thus, within the interval $[-0, 001, 0.001]$, the PDF of Z is approximately constant. Using the formula $P(z - \delta \leq Z \leq z + \delta) \approx f_Z(z) \cdot 2\delta$, with $z = 0$ and $\delta = 0.001$, we obtain

$$P(|W| < 0.001) \approx P(-0.001 \leq Z \leq 0.001) \approx f_Z(0) \cdot 0.002 = \frac{0.002}{\sqrt{2\pi}(1/\sqrt{96})} = 0.0078.$$

4. It is well known that infants born to mothers who smoke tend to be small and prone to a range of ailments. It is conjectured that also they look abnormal. Nurses were shown selections of photographs of babies, one half of whom had smokers as mothers; the nurses were asked to judge from a baby's appearance whether or not the mother smoked. In 1500 trials the correct answer was given 910 times. Is the conjecture plausible? If so, why?

   Solution: A superficially plausible argument asserts that, if all babies look the same, then the number X of correct answers in n trials is a random variable with the $\text{bin}(n, \frac{1}{2})$ distribution. Then, for large n,

$$P\left(\frac{X - \frac{1}{2}n}{\frac{1}{2}\sqrt{n}} > 3\right) \approx 1 - \Phi(3) \approx \frac{1}{1000}$$

   by the central limit theorem. For given values of n and X,

$$\frac{X - \frac{1}{2}n}{\frac{1}{2}\sqrt{n}} = \frac{910 - 750}{5\sqrt{15}} \approx 8.$$

   Now we might say that the event $\{X - \frac{1}{2}n > \frac{3}{2}\sqrt{n}\}$ is sufficiently unlikely that its occurrence casts doubt on the original supposition that babies look the same.